MIL-E-24269A(SHIPS)
4 October 1968
SUPERSEDING
MIL-E-24269(SHIPS)
18 November 1966
(See 6.3)

MILITARY SPECIFICATION

EXTINGUISHER, FIRE, CARBON DIOXIDE, 15 POUND,

PORTABLE, PERMANENT SHUTOFF,

NAVY SHIPBOARD USE

1. SCOPE

1.1 Scope. This specification covers the requirements for a 15 pound size, portable CO_2 fire extinguisher and shock proof mounting brackets.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
BB-C-101 -Carbon Dioxide (CO_2) Technical and U. S. P.
RR-C-901 -Cylinders, Compressed Gas; With Valve or Plug and Cap; 1CC 3AA.

MILITARY
MIL-S-901 -Shock Tests, H. I. (High Impact); Shipboard Machinery Equipment and Systems. Requirements for.
MIL-H-16236-Hose, Fire-Extinguisher (Portable, 15 Pound, Carbon-Dioxide).
MIL-C-16310-Cylinder, Compressed Gas (Compressed Air and Carbon Dioxide Nonshatterable and Nonmagnetic).
MIL-V-17360-Valve, Cylinder, Gas, Carbon Dioxide Fire Extinguisher.

STANDARDS

FEDERAL
FED-STD-595-Colors.

MILITARY
MIL-STD-101-Color Code for Pipeline and For Compressed Gas Cylinders.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2. Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

FSC 4210

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NATIONAL BUREAU OF STANDARDS
Handbook H-28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The fire extinguishers shall consist of a pressure cylinder, nameplate, valve assembly with siphon tube, carbon dioxide charge horn bracket, and discharge horn with hose.

3.1.1 Quick release, shock proof mounting brackets which meet the requirements of 3.11 may be required either as a part of the extinguisher supply contract or on a separate contract (see 6.2).

3.2 Material. Material shall be as specified herein or as shown on approved drawings (see 3.11).

3.3 Cylinder. Unless otherwise specified (see 6.2), the cylinder shall be in accordance with RR-C-901, type II, class 1 except as follows:

Outside diameter	- 6-3/16 inches, minimum 6-15/16 inches, maximum.
Length	- 27 inches, maximum.
Capacity	- 616 cubic inches, minimum
Weight fully charged (including valve)	- 54 pounds, maximum.
Valve connection	- 1 inch - 11-1/2 NGT female.
Marking	- 1/4 inch high characters, tare weight not required.

3.3.1 When nonmagnetic fire extinguishers are required (see 6.2), the cylinder shall comply with MIL-C-16310, size A.

3.4 Valve. The valve shall be in accordance with class b of MIL-V-17360 and shall be fitted with a heavy nonferrous copper base alloy siphon tube.

3.5 Hose. Each extinguisher shall be fitted with a discharge hose conforming to MIL-H-16236, except that the hose shall withstand a hydrostate test pressure of not less than 3,000 P.S.I. The ends of the hose shall be fitted with brass connections internally threaded as follows:

Valve discharge connection - 1/4 - 18 NPS female.
Horn connection 1/4 - 18 NPS female.

3.5.1 The length of the hose and fittings shall be approximately 37 inches.

3.6 Threads. Threads shall be in accordance with H-28.

3.7 Horn. The discharge horn shall be conical in shape and shall be fabricated of tough, rigid temperature and water-resistant material having a 1/4 - 18 NPT internal thread bronze or brass connection molded or otherwise secured to the horn for attaching the handle. Where metal fasteners are used for attaching the insert, the handle assembly shall incorporate a tight fitting sleeve of dielectric material to completely cover such fasteners. The horn material shall be capable of withstanding the test of 4.3.4. The method of attaching the handle to the horn tube shall be permanent and shall allow no separation when the extinguisher is handled or discharged. The handle shall be not less than 6 inches long. Handles may be either plastic, rubber, or seasoned hardwood. Joints of wood handles shall be glued with waterproof cement. All surfaces of wooden handles shall be thoroughly coated with a water-resistant, non-conducting enamel. The overall length of the horn and handles assembly shall be not less than 28 inches nor more than 30 inches. The horn and handle shall be black.

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3.8 Horn bracket. Each extinguisher shall be provided with a metal clip or bracket to hold the discharge horn when not in use. The bracket shall be of such design that the horn will be rigidly and securely held, yet may be readily removed without damage.

3.9 Nameplate. Each extinguisher shall be provided with a nameplate as shown on figure 1.

3.10 Charge. Unless otherwise specified (see 6.2), the standard gas charge for each extinguisher, shall be 15 pounds plus or minus 1/4 pound of carbon dioxide conforming to BB-C-101, grade B, type II.

3.10.1 When specified (see 6.2), the undercharge for an extinguisher intended for use when the temperature is 130°F. or higher, shall be 90 percent of 15 pounds, or 13-1/2 pounds plus or minus 1/4 pound of carbon dioxide as specified in 3.10.

3.11 Extinguisher holder device. The bracket for mounting the extinguisher to a bulkhead, shall consist of a hook for the valve hanger lug and a clamping band with a quick release device which can be easily operated with one hand. The location of the hook must be adjustable so that no strain will be applied to the valve body by tightening of the clamping band. The length of the clamping band shall be adjustable to securely hold any extinguisher cylinder meeting the dimensional requirements of 3.3 when tested in accordance with 4.3.5. A detail drawing of the bracket shall be submitted to Naval Ship Engineering Center for approval prior to manufacture or tests.

3.11.1 Brackets for mounting nonmagnetic fire extinguishers (see 3.3.1) shall be fabricated of nonmagnetic materials having a magnetic permeability of not over 2.0 in the finished form. The brackets shall be adjustable to fit any cylinder meeting the dimensional requirements of MIL-C-16310 in a manner equal to the above.

3.12 Cleaning. After final assembly, and before painting, the extinguishers shall be thoroughly cleaned for all dirt, oil or other foreign material. Burrs and sharp edges shall be removed.

3.13 Painting. Cylinders shall be painted with color No. 11105 red gloss conforming to FED-STD-595. The valve and valve handle shall not be painted.

3.13.1 Unless otherwise specified (sec 6.2), extinguishers shall be stenciled in accordance with MIL-STD-101.

3.14 Instruction cards. In addition to the nameplate band required in 3.9, the manufacturer shall furnish with each extinguisher a set of instructions for operating and recharging the extinguisher. Instruction cards shall be approved by the contracting officer.

3.15 Each extinguisher shall be provided with a spare safety disc and washer enclosed in a plastic envelope secured to the neck of the cylinder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Complete records of the inspection, examination and tests shall be kept by the supplier and made available to the Government representative as specified in the contract or order.

4.2 Examination. Each extinguisher offered for delivery on a contract or order shall be examined to determine conformance to the requirements of this specification as outlined below. Failure to pass any examination or noncompliance with the following shall be cause for rejection.

(a) Dimensions and weight in accordance with 3.3.
(b) Presence of all parts and components.
(c) Completeness and correctness of assembly.

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(d) Safety devices and guards.
(e) Suitability of material and workmanship.
(f) Finish - defects such as scale, lumps, burns, cracks, thin spots, color not as specified or streaked and uneven.
(g) Presence and acceptability of spare parts, accessories, identification markings and other required operating and safety instructions.

4.3. Tests.

4.3.1 Leakage test. Each fully charged extinguisher shall be submerged in water at a temperature of not less than 90°F. nor more than 110°F. for a period of 1 hour. The horn and hose may be removed for this test. During the test each extinguisher shall be inspected for leaks. Other methods of leak detection may be used subject to approval of the Naval Ship Engineering Center prior to issue of invitation for bid. Any alternate method shall have a sensitivity capable of detecting a leak rate of one ounce of carbon dioxide per year. Any signs of escape of CO_2 gas from the cylinder or valve shall be cause for rejection. Following completion of testing, all traces of moisture shall be removed from the operating portion of the valve and the safety-release mechanism.

4.3.2 Cylinders. Cylinders shall be inspected in accordance with RR-C-901 or MIL-C-16310, as applicable.

4.3.3 Valves. Valves shall be inspected in accordance with MIL-V-17360.

4.3.4 Test of horn. One horn shall be selected at random from each 500 horns manufactured for Navy use, and shall be subjected, in turn, to each of the following tests in sequence. At the conclusion of the tests, the horn shall be examined for cracks, distortion, or other damage. If the test horn fails in any manner, all horns from the lot from which the test horn was selected shall be tested.

4.3.4.1 Impact shock. The test horn shall be firmly supported by a clamping mechanism at the handle so that the point of impact will be on the external surface approximately 2 inches from the large end of the horn. The horn shall then be connected to a fully charged extinguisher and the extinguisher shall be discharged until the end of the snow discharge. A 5-pound steel ball shall immediately be dropped, free and unguided, upon the horn from a height of 2 feet. This test shall be repeated a total of four times.

4.3.4.2 Thermal shock. The test horn shall be soaked in an ice brine bath at minus 25°F. for a period of 30 minutes, then removed rapidly and plunged into boiling water for a period of 30 seconds. Fractures or cracks shall be basis for rejection.

4.3.4.3 Accelerated aging. The horn shall be placed in a pressure bomb, pressurized with oxygen at 290 to 310 p.s.i. and then placed in an oven. Heat to 160°F. and hold at this temperature for 24 hours. Let the bomb and test horn cool, then release pressure. Remove and subject horn to impact test of 4.3.4.1.

4.3.4.4 Dielectric test. An A.C. test potential of 5000 volts shall be applied to a 1/8-inch thick test sample of horn material for a period of 1 minute. Arcing or evidence of electric breakdown shall be considered evidence of failure and cause for rejection.

4.3.5 Shock test. A mounting bracket with fire extinguisher shall be mounted on a lightweight shock machine and tested in accordance with MIL-S-901, grade A, class 1. For this test the cylinder shall contain 15 pounds of water and shall be pressurized to 870 p.s.i. with air or nitrogen. Changes in bracket design subsequent to this test shall require retest unless a specific waiver is granted by Naval Ship Engineering Center.

4.3.5.1 Certification. The supplier shall furnish written certification with each contract, that he has determined (either by actual test, or through adequate control of the manufacturing process), that each extinguisher and bracket meets the requirements of this specification. Certification shall contain actual test, examination or other verifiable quality data.

4.4 Inspection of preparation for delivery. Packaging, packing, and marking shall be inspected to assure compliance with section 5 of this specification.

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5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements.)

5.1 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

5.2 Domestic shipment and early equipment installation.

5.2.1 Preservation and packaging. Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.2.2 Packing. Packing shall be accomplished in a manner which will insure acceptance to common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.2.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.3 Domestic shipment and storage or overseas shipment. The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (See 6.2).

5.3.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

(5.3.1.1 Preservation and packaging. Preservation and packaging shall be Level A or C, as specified (see 6.2).

5.3.1.1.1 Level A.

5.3.1.1.1.1 Unit containers. Unit containers, of the extinguishers, shall be in accordance with method III of MIL-P-116. The unit package shall contain one extinguisher in a fiber board box in accordance with PPP-B-636, class weather resistant, type SF.

5.3.1.1.2 Level C.

5.3.1.1.2.1 Unit Containers. Unit containers of the extinguishers shall be in accordance with the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.3.1.2 Packing. Packing shall be Level A, B, or C, as specified (see 6.2).

5.3.1.2.1 Level A.

5.3.1.2.1.1 Exterior containers. Extinguishers, packaged as specified (see 6.2), shall be packed in containers conforming to PPP-B-640, class 2. All center and edge seams and manufacturer's joints are to be sealed and waterproofed with pressure sensitive tape in accordance with the applicable fiberboard box specification. Shipping containers shall be closed, strapped, or banded in accordance with the applicable box specification or appendix thereto. The gross weight shall not exceed the weight limitations of the applicable box specification.

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5.3.1.2.2 Level B.

5.3.1.2.2.1 Exterior containers. Extinguishers, packaged as specified (see 6.2), shall be packed in containers conforming to PPP-B-640, class I. Shipping containers shall be closed in accordance with the applicable box specification or appendix thereto. The gross weight shall not exceed the weight limitations of the applicable specification.

5.3.1.2.3 Level C.

5.3.1.2.3.1 Exterior containers. Extinguishers, packaged as specified (see 6.2), shall be packed in containers, in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the first using activity for early installation. This level in general shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice.

5.3.1.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked for shipment in accordance with MIL-STD-129.)

5.4 Use of polystyrene (loose-fill) material.

5.4.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.2), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labelled as follows:

" CAUTION

Contents cushioned etc. with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber,
fiberboard or transparent flexible cellular material."

5.4.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler and dunnage.

6. NOTES

6.1 Intended use. These portable CO_2 fire extinguishers are intended for use in locations where class B and class C fires are anticipated; that is, where there is a fire hazard involving oils, greases, flammable volatile liquids, or energized electrical equipment (since the gas from CO_2 extinguishers combats vapor-phase combustion and is a nonconductor of electricity).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Number of extinguishers and brackets required (see 3.1).
(c) Cylinder, if different (see 3.3).
(d) Nonmagnetic cylinders, if required (see 3.3.1).
(e) Standard gas charge, if other than that specified (see 3.10).
(f) When undercharge for extinguisher is required (see 3.10.1)
(g) When stenciling is required (see 3.13.1)
(h) Preservation, packaging, packing, or marking requirements other than those required by 5.2 (see 5.3).
(i) When use of polystyrene is permitted (see 5.4).

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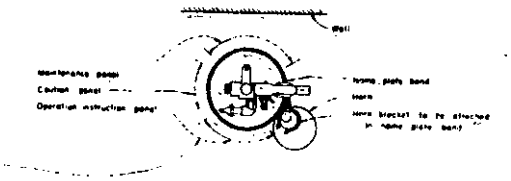
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GENERAL NOTES:

1. Material – Brass or CRES 0.020" thick (min).
2. Letters shall be of two sizes, 1/8" and 3/16" high, as shown. The letters shall be deep etched and filled with black enamel.
3. Polish and apply one coat of clear lacquer after letters have been filled in.
4. Length and type of fastening shown are descriptive only and may vary, depending on diameter of the cylinder and method of fastening strap.
5. General arrangement of name plate and horn clip on Military CO_2 extinguishers as indicated.



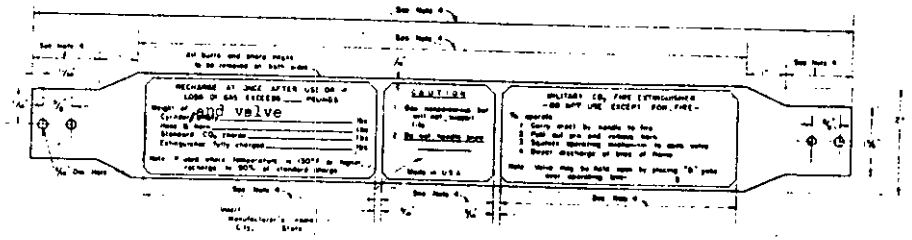


Figure 1 - Nameplate band for Military CO_2 fire extinguisher.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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